```
"C:\Users\nicop\OneDrive\Documentos\Universidad\Proyecto de Grado STREDEMIA\STREDEMIA\.venv\Scripts\python.exe" "C:\Users\nicop\OneDrive\Documentos\Universidad\Proyecto de Grado STREDEMIA\STREDEMIA\.venv\Modelos\DNN_Model.py"
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1776291679.666761    2856 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1776291682.713656    2856 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
Entrenamiento: (381, 33)
Validación:    (37, 33)
Prueba:        (37, 33)
Variables de entrada (features): 33
  - Ítems SISCO:     21
  - Variables contextuales: 12
Scaler guardado en carpeta 'Modelos'

inicio de validación cruzada con 5 folds (pliegues)...

*-*-*-* Fold 1/5
I0000 00:00:1776291684.660918    2856 cpu_feature_guard.cc:227] This TensorFlow binary is optimized to use available CPU instructions in performance-critical operations.
To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags.
WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for TensorFlow >= 2.11. Even if CUDA/cuDNN are installed, GPU will not be used. Please use WSL2 or the TensorFlow-DirectML plugin.
F1-score en fold 1: 0.9647
*-*-*-* Fold 2/5
F1-score en fold 2: 0.9172
*-*-*-* Fold 3/5
WARNING:tensorflow:5 out of the last 7 calls to <function TensorFlowTrainer.make_predict_function.<locals>.one_step_on_data_distributed at 0x00000216C510B880> triggered tf.function retracing. Tracing is expensive and the excessive number of tracings could be due to (1) creating @tf.function repeatedly in a loop, (2) passing tensors with different shapes, (3) passing Python objects instead of tensors. For (1), please define your @tf.function outside of the loop. For (2), @tf.function has reduce_retracing=True option that can avoid unnecessary retracing. For (3), please refer to https://www.tensorflow.org/guide/function#controlling_retracing and https://www.tensorflow.org/api_docs/python/tf/function for  more details.
WARNING:tensorflow:6 out of the last 9 calls to <function TensorFlowTrainer.make_predict_function.<locals>.one_step_on_data_distributed at 0x00000216C510B880> triggered tf.function retracing. Tracing is expensive and the excessive number of tracings could be due to (1) creating @tf.function repeatedly in a loop, (2) passing tensors with different shapes, (3) passing Python objects instead of tensors. For (1), please define your @tf.function outside of the loop. For (2), @tf.function has reduce_retracing=True option that can avoid unnecessary retracing. For (3), please refer to https://www.tensorflow.org/guide/function#controlling_retracing and https://www.tensorflow.org/api_docs/python/tf/function for  more details.
F1-score en fold 3: 0.9409
*-*-*-* Fold 4/5
F1-score en fold 4: 0.8928
*-*-*-* Fold 5/5
F1-score en fold 5: 0.9643

*-*-*-* Resumen validación cruzada *-*-*-*-*
  Fold 1: F1-macro = 0.9647
  Fold 2: F1-macro = 0.9172
  Fold 3: F1-macro = 0.9409
  Fold 4: F1-macro = 0.8928
  Fold 5: F1-macro = 0.9643
  Promedio: 0.9360
  Desv. estándar: 0.0278
```

```

*-*-*-*- Entrenamiento de modelo final *-*-*-*-*-
Epoch 1/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 1s 19ms/step - accuracy: 0.4785 - loss: 1.0277 - val_accuracy: 0.4054 - val_loss: 1.0002
Epoch 2/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.6388 - loss: 0.8204 - val_accuracy: 0.4595 - val_loss: 0.8669
Epoch 3/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.7297 - loss: 0.6145 - val_accuracy: 0.6757 - val_loss: 0.7216
Epoch 4/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.7656 - loss: 0.5023 - val_accuracy: 0.6757 - val_loss: 0.6475
Epoch 5/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.8254 - loss: 0.4135 - val_accuracy: 0.7027 - val_loss: 0.5570
Epoch 6/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.8660 - loss: 0.3407 - val_accuracy: 0.7568 - val_loss: 0.4737
Epoch 7/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.8876 - loss: 0.3079 - val_accuracy: 0.7568 - val_loss: 0.4637
Epoch 8/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.8900 - loss: 0.2728 - val_accuracy: 0.7838 - val_loss: 0.4556
Epoch 9/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.9139 - loss: 0.2448 - val_accuracy: 0.8378 - val_loss: 0.4016
Epoch 10/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - accuracy: 0.9115 - loss: 0.2302 - val_accuracy: 0.8649 - val_loss: 0.3272
Epoch 11/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9211 - loss: 0.2115 - val_accuracy: 0.9189 - val_loss: 0.3256
Epoch 12/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.9139 - loss: 0.2014 - val_accuracy: 0.9189 - val_loss: 0.3169
Epoch 13/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9378 - loss: 0.1690 - val_accuracy: 0.8919 - val_loss: 0.2594
Epoch 14/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.9282 - loss: 0.1887 - val_accuracy: 0.9189 - val_loss: 0.2416
Epoch 15/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9474 - loss: 0.1485 - val_accuracy: 0.9459 - val_loss: 0.2082
Epoch 16/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.9258 - loss: 0.1720 - val_accuracy: 0.9459 - val_loss: 0.2049
Epoch 17/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.9450 - loss: 0.1424 - val_accuracy: 0.9730 - val_loss: 0.1670
Epoch 18/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9569 - loss: 0.1391 - val_accuracy: 0.9730 - val_loss: 0.1720
Epoch 19/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9617 - loss: 0.1068 - val_accuracy: 0.9730 - val_loss: 0.1091
Epoch 20/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.9665 - loss: 0.1057 - val_accuracy: 0.9459 - val_loss: 0.1515
Epoch 21/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - accuracy: 0.9617 - loss: 0.1004 - val_accuracy: 0.9459 - val_loss: 0.1401
Epoch 22/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.9641 - loss: 0.0877 - val_accuracy
```

```
: 0.9730 - val_loss: 0.1219
Epoch 23/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9450 - loss: 0.1305 - val_accuracy: 0.9730 - val_loss: 0.1307
Epoch 24/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9617 - loss: 0.0842 - val_accuracy: 0.9459 - val_loss: 0.1618

*-*-*-*- Evaluación final sobre datos de prueba *-*-*-*-

Reporte de clasificación:
              precision    recall  f1-score   support

        bajo       1.00      1.00      1.00         3
       medio       0.96      0.93      0.95        28
        alto       0.71      0.83      0.77         6

    accuracy                           0.92        37
   macro avg       0.89      0.92      0.90        37
weighted avg       0.93      0.92      0.92        37

Modelo entrenado con éxito...

*-*-*-* MÉTRICAS Y EVALUACIONES *-*-*-*-*

AUC-ROC
Macro (one-vs-rest): 0.9406

Por clases:
  AUC-ROC bajo: 1.0000
  AUC-ROC medio: 0.8810
  AUC-ROC alto: 0.9409
Curvas ROC por clases archivadas en carpeta 'Resultados'
Matriz de confusión almacenada en carpeta 'Resultados'
Curvas de entrenamiento guardadas en carpeta 'Resultados'

Modelo ejecutado con éxito.
Métricas y evaluaciones generadas con éxito
Modelo Guardado (Carpeta Modelos)

Process finished with exit code 0
```